CompanyIQ — LLM Quality vs. Cost Comparison for the Company-Analysis Task

Date: 12 June 2026 **Scope:** Empirical benchmark of every model wired into CompanyIQ's provider registry, evaluated on the *actual* production scoring step (binary measure scoring against a framework, with verbatim-evidence requirements and strict anti-inference rules).

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1. Why a task-specific comparison (not generic benchmarks)

CompanyIQ does not ask a model to "be smart" in the abstract. It asks each model to perform one narrow, high-stakes job, repeated tens of thousands of times:

> Given a measure definition, scoring guidance, and a pack of retrieved evidence text, decide **Yes / No / Partial**, assign a **confidence**, and return **verbatim quotes** that are actually present in the evidence — while obeying anti-inference rules (e.g. alliance membership is not a company commitment; intensity targets are not absolute targets).

For this job, the qualities that matter are not the ones generic leaderboards reward. They are, in order:

1. **Evidence discipline** — does the model ground its verdict in quotes that are genuinely in the source text, or does it paraphrase/fabricate?
2. **Anti-inflation / strictness** — does it resist scoring "Yes" on weak or inferential evidence? (Per the project's quality-control preference, the risk is *inflated* scores, not missed ones.)
3. **Verdict stability / agreement** — does it agree with the cross-model consensus on the same evidence, or is it an outlier?
4. **Reliability under batch load** — does it complete every call without rate-limit or format failures across a long batch?
5. **Cost per company** — total spend to score a full framework for one company.
6. **Latency** — wall-clock per measure (matters for batch throughput).

This document scores every model on exactly these axes, using a replay of the production prompt on real companies from the live database.

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2. Method

A standalone harness (`server/scripts/benchmark.ts`) imports the **real** production functions — the evidence-pack builder (`buildEvidencePacksForCategory`, BM25 retrieval, 1,500-char chunking, 8,000-char cap) and the **exact** binary-scoring system/user prompt from `analyzer.ts` — and replays the identical scoring call across all candidate models. The only variable is the model; evidence and prompt are byte-identical per cell.

Parameter	Value
Frameworks	#7 AI Governance & Strategy (34 measures), #3 Bank Financed Emissions (27 measures)
Companies	Capgemini, Tyson Foods, Delta Electronics (AI Gov); KBC Groep, AlRayan Bank (Bank Emissions)
Measures sampled	8 per framework (evenly across categories)
Evidence source	Live production DB — real fetched documents, deduplicated via `document_content`
Cells	40 unique (company × measure) cells
Calls	40 × 9 models = **360 real scoring calls**
Decoding	`temperature: 0`, `maxTokens: 2000`, JSON mode where supported

Quote-grounding metric: each returned quote is checked for verbatim presence in the evidence text (exact match, with a 60%-contiguous-window fallback). The grounding rate is the fraction of quotes that are genuinely traceable to the source — a direct, objective measure of evidence discipline.

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3. Headline results

All metrics computed over the 360 calls. “Quality on success” excludes failed calls so a reliability problem (rate-limits) does not get mislabeled as a quality problem.

Model	Route	Completion	Quote grounding	Consensus agreement	Yes-rate (strictness)	Avg latency	Cost / company (34 meas)
DeepSeek V4-Flash (`deepseek-chat`)	DeepSeek direct	**100%**	**97%**	**100%**	15%	4.2 s	**$0.015**
DeepSeek V3.1	OpenRouter	**100%**	**97%**	95%	10%	17.7 s	$0.029
DeepSeek V4-Pro	DeepSeek direct	92%	94%	**97%**	5%	18.2 s	$0.046
Claude Sonnet 4.5	Anthropic direct	**100%**	95%	95%	20%	11.6 s	$0.517
GPT-4o	OpenAI direct	**100%**	92%	95%	15%	2.9 s	$0.308
MiniMax-Text-01	MiniMax direct	**100%**	92%	88%	28%	12.8 s	$0.036
Mistral Large	Mistral direct	95%	92%	92%	24%	6.4 s	$0.250
DeepSeek R1-0528	OpenRouter	85%	86%	88%	3%	48.8 s	$0.070
Gemini 2.5 Flash	Google direct	**18%**	64%	100%	29%	1.3 s	$0.052

Quality vs Cost scatter

Notes:

- **Cost** uses measured token footprint (~2,746 input + 180–460 output tokens per measure) × current per-million pricing (sourced June 2026; see §6).
- **Gemini's 18% completion** is entirely **rate-limit / quota exhaustion** (33 of 40 calls returned HTTP 429 "quota exceeded" or "high demand"), not a quality defect. On its 7 successful calls its grounding was weaker (64%) and it leaned permissive (high Yes-rate). This empirically reproduces the known batch-reliability issue that already drove the "DeepSeek-primary" preference.
- **DeepSeek R1** failures were format-related (5 JSON-parse failures from reasoning leakage, 1 other) even after <think>-stripping and a JSON-only retry.

- **DeepSeek V4-Pro** had 3 JSON-parse failures (of 40) on the most evidence-dense measures — as a "thinking" model it occasionally leaks reasoning into the output. This is milder than R1 and is addressable with the same defensive parsing the analyzer already applies.

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4. What the numbers mean for *this* task

Evidence discipline (the most important axis)

DeepSeek V4-Flash and DeepSeek V3.1 lead on verbatim grounding (**97%**), with Claude just behind (95%). In practice this means when these models cite a quote, it is almost always genuinely in the source — exactly what an auditable assessment needs. Gemini's 64% means roughly a third of its quotes could not be located verbatim, a real provenance risk for a sourced framework.

Strictness and score inflation

The frameworks are deliberately hard (low Yes-rates are expected and desirable). Reading the Yes-rate as a strictness dial:

- **R1 is the strictest** (3% Yes) — but this cuts both ways (see false-negative finding below).
- **DeepSeek V4-Pro is the next strictest (5% Yes) — and, crucially, its strictness is well-calibrated, not over-rejection** (see below). This makes it the standout "high-tier" candidate.
- **DeepSeek V4-Flash, GPT-4o** sit at a healthy 15%; **Claude** at 20%.
- **MiniMax (28%) and Mistral (24%)** are the most permissive — a mild score-inflation tendency, which is the specific failure mode the project's QC preference warns against.

Verdict stability

40 cells, 8 models: **33 of 40 (≈83%) were unanimous**, which is strong evidence the framework + prompt are robust regardless of model. The 7 split cells were all genuinely judgment-heavy measures. Two concrete patterns from the raw data:

- **R1 over-rejects on present evidence.** On KBC's *coal-capacity-exclusion* and *sectoral-lending-exposure* measures, R1 was the **lone "No" against six "Yes"** — its extreme strictness produces false negatives, not just resistance to inflation.
- **MiniMax/Mistral over-accept on borderline measures.** They appear on the "Yes" side of nearly every split (e.g. Capgemini *strategic-AI-partnerships*, Delta *AI-reskilling*), consistent with their higher Yes-rates.

DeepSeek V4-Flash matched the consensus on **100%** of cells — the best stability of any model tested.

DeepSeek V4-Pro: strict but well-calibrated (the key new finding)

Unlike R1, V4-Pro's higher strictness is *discriminating*, not *over-rejecting*. Across its 37 successful cells it:

- diverged from consensus on **only 1 cell** (Capgemini *AI-reskilling-programme*, where it said "No" against four "Yes"),
- **never** over-accepted (0 cells where it said "Yes" while the majority said "No"), and

- split from the cheaper V4-Flash on **only 1 of 40 cells**.

In other words, V4-Pro reaches essentially the same verdicts as V4-Flash but is slightly more conservative on genuinely borderline measures — the desirable direction for a framework whose main risk is score inflation. Its one weakness is reliability: 3 JSON-parse failures (92% completion) on the hardest measures, versus V4-Flash's flawless 100%.

Reliability under batch load

This is decisive for CompanyIQ's batch workload. Only five models completed 100% of calls under concurrent load: **DeepSeek V4-Flash, DeepSeek V3.1, Claude, GPT-4o, MiniMax**. Gemini collapsed to 18% on quota; R1 dropped to 85% on format; Mistral had 2 transient 429s.

Latency

GPT-4o (2.9 s) and DeepSeek V4-Flash (4.2 s) are fastest. R1 is by far the slowest (48.8 s avg, 128 s p90) — unsuitable as a default batch backend regardless of its strictness appeal.

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5. Cost in context

Per **full company analysis** (34 measures, single pass):

Tier	Models	Cost/company	Relative to DeepSeek
Ultra-low	DeepSeek V4-Flash $0.015, V3.1$0.029, MiniMax $0.036	<$0.04	1–2.4×
Low	DeepSeek V4-Pro $0.046, Gemini$0.052, R1 $0.070	<$0.08	3–4.7×
Premium	Mistral $0.250, GPT-4o$0.308, **Claude $0.517**	$0.25–0.52	17–35×

At batch scale this dominates the economics. Scoring the full corpus (≈2,565 companies × 34 measures) once:

Model	Approx. batch cost (one pass, all companies)
DeepSeek V4-Flash	**≈ $38**
DeepSeek V3.1 (OpenRouter)	≈ $75
MiniMax	≈ $92
DeepSeek V4-Pro	≈ $118
GPT-4o	≈ $790
Claude Sonnet 4.5	**≈ $1,325**

DeepSeek V4-Pro is notable here: it delivers near-Claude-level conservative judgment and the second-best consensus agreement (97%) at **~11× less cost** than Claude and only ~3× more than V4-Flash.

Claude delivers marginally lower grounding (95% vs 97%) and **equal** consensus agreement (95% vs 100%) versus DeepSeek V4-Flash, at **~35× the cost**. For this task, the premium does not buy measurably better evidence discipline.

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6. Pricing sources (per 1M tokens, cache-miss input / output, June 2026)

Model	Input	Output	Source
DeepSeek V4-Flash (`deepseek-chat`)	$0.14	$0.28	api-docs.deepseek.com/quick_start/pricing
DeepSeek V4-Pro	$0.435	$0.87	api-docs.deepseek.com/quick_start/pricing
DeepSeek R1-0528	$0.50	$2.15	openrouter.ai/api/v1/models
DeepSeek V3.1	$0.21	$0.79	openrouter.ai/api/v1/models
Claude Sonnet 4.5	$3.00	$15.00	anthropic.com/news/claude-sonnet-4-5
GPT-4o	$2.50	$10.00	openai.com/api/pricing
Gemini 2.5 Flash	$0.30	$2.50	ai.google.dev/gemini-api/docs/pricing
Mistral Large	$2.00	$6.00	openrouter.ai (mistral-large)
MiniMax (M2 ref)	$0.26	$1.00	openrouter.ai (minimax-m2)

> **Action item flagged:** DeepSeek is migrating to **V4**; the legacy `deepseek-chat` / `deepseek-reasoner` model names are **deprecated 2026-07-24** (they now map to V4-Flash non-thinking/thinking). The direct `deepseek` provider should be updated to the V4 names before that date.

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7. Recommendation

For CompanyIQ's evidence-based company-analysis task, the data supports a clear tiering that also satisfies the multi-model redundancy preference:

Role	Model	Rationale
Primary (batch default)	**DeepSeek V4-Flash**	Best evidence grounding (97%) and verdict stability (100% consensus), 100% completion under load, fastest-tier latency, cheapest by a wide margin. Already the project's chosen primary; the data strongly validates it.
High-tier quality option	**DeepSeek V4-Pro**	The best new addition. Strict but well-calibrated (5% Yes, 0 over-accepts, 97% consensus), 94% grounding, ~11× cheaper than Claude. Use it as the rigorous "second opinion" / arbiter on borderline measures, or as a higher-conviction batch backend where the 92% completion (fixable parsing) is acceptable. A far better high-tier DeepSeek choice than R1 (4× faster, better calibrated).
Backup #1 (premium cross-check)	**Claude Sonnet 4.5**	Highest-tier reasoning, 95% grounding, 100% completion. Use as the arbiter on split measures or spot-check QA — not the batch default (35× cost).
Backup #2 (independent route, low cost)	**DeepSeek V3.1 via OpenRouter**	Matches V4-Flash on grounding (97%), independent network path (resilience if DeepSeek-direct rate-limits), still ultra-cheap.
Tertiary fallback	**GPT-4o**	Fast, reliable, 92% grounding; a fully independent vendor for redundancy.
Use with caution	MiniMax, Mistral	Functional now (after the fixes below) but mildly inflation-prone (higher Yes-rate); acceptable as overflow capacity, not for primary scoring.
Not recommended as default	Gemini 2.5 Flash (rate-limits under batch), DeepSeek R1 (too slow at ~49 s/call; over-rejects valid evidence; format instability)	V4-Pro now supersedes R1 as the strict high-tier DeepSeek option.

Optional high-confidence mode: for the subset of judgment-heavy measures, an **ensemble** of DeepSeek V4-Flash + DeepSeek V4-Pro (with disagreement escalated to Claude) gives a strict, well-calibrated cross-check for only ~$0.06/company — essentially free relative to a Claude-only pass, while keeping a premium arbiter only for the rare splits.

8. Provider-wiring fixes made during this work

These were discovered empirically while building the benchmark and are now fixed in `server/lib/ai-providers.ts`:

1. **Mistral was non-functional.** Every Mistral scoring call returned HTTP 422 because the provider sent a `seed` parameter that the Mistral API rejects (`extra_forbidden`). Added a `supportsSeed` flag (default true) and disabled it for Mistral. Mistral now scores correctly.
2. **MiniMax base URL + JSON mode.** Corrected base URL to `https://api.minimax.io/v1` (the old `api.minimax.chat` returns 401) and disabled `response_format: json_object` (MiniMax rejects it). MiniMax now scores correctly.
3. **OpenRouter added** as two providers — `openrouter` (DeepSeek V3.1) and `deepseek-r1` (DeepSeek R1-0528) — with key rotation and the required OpenRouter attribution headers, and a `supportsJsonMode` opt-out for reasoning models.

All four changes are backward-compatible and isolated to the provider layer.